Signed Surface-Envelope Transductively Standardized Conformal Prediction

Construction, finite-sample proofs, and comparison with the separated shortcut

Method development note

September 8, 2026

Abstract

We derive a surface-envelope refinement of globally worst-case (GWC) transductive standardization. The construction allows signed residuals, retains the location and scale in a single ratio, and produces coordinatewise residual thresholds without enumerating a multidimensional grid. Its coverage follows from deterministic containment of the full conformal set. For nonnegative scores and the same residual representation, its rectangle is contained in the old separated mean-row shortcut, with the qualifications and endpoint conventions stated below. This is weak, samplewise improvement of region size, not strict improvement on every sample, a comparison of coverage probabilities, or a uniform comparison with CQHR. We give an unconditional backward-search algorithm and a proved range on which binary search is valid.

Contents

1 Setup and the conformal rank

Condition throughout on an independently trained predictor and any training-only choices. Let $(X^i, Y^i)_{i=1}^{n+1}$ be exchangeable and let

$$\mathbf{E}^i = \big(E_1(X^i, Y_1^i), \dots, E_d(X^i, Y_d^i)\big) \in \mathbb{R}^d$$

be coordinatewise nonconformity scores. The first n residual vectors are observed. Fix a test input x and let $\mathbf{e} = \mathbf{E}(x, y)$ denote a candidate residual. Suppose its attainable coordinate values lie in

$$D_j(x) = [a_j(x), \infty) \cap \mathbb{R}, \qquad D(x) = \prod_{j=1}^{d} D_j(x), \qquad a_j(x) \in \mathbb{R} \cup \{-\infty\}.$$

Here $a_j(x)$ is a known bound for the test residual, not an estimated bound on the distribution. Calibration residuals at other covariates need not exceed $a_j(x)$. A larger enclosing domain, including $D_j = \mathbb{R}$, is also valid. Coordinatewise score sublevel sets must be intervals for the final prediction set in outcome space to be a rectangle; validity itself does not require that.

Write $N = n + 1$ and

$$r = \lceil N(1-\alpha) \rceil, \qquad \mathcal{Q}_\alpha(v_1, \dots, v_n) = \begin{cases} v_{(r)}, & r \le n, \\ +\infty, & r = N. \end{cases} \tag{1}$$

Use weak acceptance inequalities throughout. No jitter is needed, and ties are allowed. If $v_i \le w_i$ for every i, then $\mathcal{Q}_\alpha(v) \le \mathcal{Q}_\alpha(w)$: for each real c, at least as many v_i as w_i are at most c, so the ordered values have the same ordering. This elementary monotonicity is used repeatedly below.

Lemma 1.1 (Exchangeable rank). *If $S^1, \dots, S^N$ are exchangeable real scores, then* $\Pr\{S^N \le \mathcal{Q}_\alpha(S^1, \dots, S^n)\} \ge 1 - \alpha$.

Proof. Randomly break ties using independent continuous auxiliary variables. The rank of the last score among the N scores is uniform on $\{1, \dots, N\}$ by exchangeability. A rank at most r implies the weak acceptance event in the statement, including when tied values occur. Its probability is therefore at least $r/N \ge 1 - \alpha$. If $r = N$, acceptance is automatic by (1). □

This is the standard conformal rank argument; see [1]. The derivations of the envelope and its comparisons below are supplied here.

2 Augmented standardization and its self-score link

For each coordinate define

$$m_j = \frac{1}{n} \sum_{i=1}^{n} E_j^i, \qquad s_j^2 = \frac{1}{n} \sum_{i=1}^{n} (E_j^i - m_j)^2. \tag{2}$$

The divisor in s_j^2 is n, whereas the variance of the augmented N-point sample below has divisor $N - 1 = n$. This distinction fixes all constants in the link. Initially assume $s_j > 0$ for every j; Section 8 supplies a valid extension to degenerate calibration samples. Finite population moments are unnecessary for the finite-sample argument.

For a candidate coordinate z, put $u = z - m_j$. The augmented mean is

$$\mu_j(z) = \frac{nm_j + z}{N} = m_j + \frac{u}{N}. \tag{3}$$

Using $\sum_i (E_j^i - m_j) = 0$ gives the full variance expansion

$$\sum_{i=1}^{n} (E_j^i - \mu_j(z))^2 + (z - \mu_j(z))^2 = \sum_{i=1}^{n} \left(E_j^i - m_j - \frac{u}{N} \right)^2 + \left(\frac{nu}{N} \right)^2$$
$$= ns_j^2 + \frac{nu^2}{N^2} + \frac{n^2 u^2}{N^2} = ns_j^2 + \frac{nu^2}{N}.$$

Consequently

$$\sigma_j(z) = \sqrt{s_j^2 + \frac{(z - m_j)^2}{N}}. \tag{4}$$

The basic coordinate transformation, its scalarization, and its candidate self-score are respectively

$$F_j(t, z) = \frac{t - \mu_j(z)}{\sigma_j(z)} = \frac{t - m_j - (z - m_j)/N}{\sqrt{s_j^2 + (z - m_j)^2/N}}, \tag{5}$$

$$\Phi_{\mathbf{z}}(\mathbf{t}) = \max_{j \le d} F_j(t_j, z_j), \tag{6}$$

$$g_j(e) = F_j(e, e) = \frac{n(e - m_j)/N}{\sqrt{s_j^2 + (e - m_j)^2/N}}. \tag{7}$$

Unlike a calibration-only fitted standardization, at the true test residual these parameters are symmetric functions of all N residual vectors.

Lemma 2.1 (Signed inverse)**.** *Let $T = n/\sqrt{N}$. The map $g_j : \mathbb{R} \to (-T, T)$ is strictly increasing, and for finite e,*

$$g_j(e) \le c \quad \Longleftrightarrow \quad e \le \omega_j(c), \tag{8}$$

where extended-real endpoints encode empty and unrestricted sublevels, and

$$\omega_j(c) = \begin{cases} -\infty, & c \le -T, \\ m_j + \dfrac{N s_j c}{\sqrt{n^2 - N c^2}}, & -T < c < T, \\ +\infty, & c \ge T. \end{cases} \tag{9}$$

Proof. Direct differentiation gives $g_j'(e) = (n/N) s_j^2 \{ s_j^2 + (e - m_j)^2/N \}^{-3/2} > 0$. Its limits at $\pm\infty$ are $\pm T$, never attained at finite e. For $|c| < T$, the equation $g_j(e) = c$, with $u = e - m_j$, yields

$$c\sqrt{s_j^2 + u^2/N} = nu/N, \quad (n^2 - Nc^2)u^2 = N^2 s_j^2 c^2, \quad \operatorname{sign}(u) = \operatorname{sign}(c).$$

Thus $u = N s_j c / \sqrt{n^2 - Nc^2}$. Keeping the sign condition is essential: squaring alone would introduce a spurious root for negative c. □

The signed residual sublevel is $D_j(x) \cap (-\infty, \omega_j(c)]$. There is no $\max\{0, \omega_j(c)\}$ in the signed link. If the upper endpoint is below a finite $a_j(x)$, the sublevel is empty.

3 The full conformal reference set

For a candidate $\mathbf{e}$ define

$$Q_{\mathbf{e}} = \mathcal{Q}_\alpha\big(\Phi_{\mathbf{e}}(\mathbf{E}^1), \dots, \Phi_{\mathbf{e}}(\mathbf{E}^n)\big), \qquad C_{\text{full}}(x) = \{y : \Phi_{\mathbf{E}(x,y)}(\mathbf{E}(x,y)) \le Q_{\mathbf{E}(x,y)}\}. \tag{10}$$

Equivalently, y is accepted if $e_j \le \omega_j(Q_{\mathbf{e}})$ for every j. The threshold depends on the entire candidate, so this reference set need not be rectangular or cheap to compute.

Theorem 3.1 (Full conformal coverage)**.** *Under exchangeability,* $\Pr\{Y^{n+1} \in C_{\text{full}}(X^{n+1})\} \ge 1-\alpha$.

Proof. At $\mathbf{e} = \mathbf{E}^{n+1}$, augmenting the calibration sample gives precisely the full exchangeable sample. The means and sample standard deviations of that sample are invariant to permutation. Applying the same standardized maximum to each of the N observations therefore gives exchangeable scalar scores. Lemma 1.1 proves the assertion. One need not assume that the scores are independent of the fitted transformation. $\square$

4 Exact coordinate suprema and signed GWC

For any nonempty interval $I = [\ell, u] \cap \mathbb{R}$, allowing infinite endpoints, define

$$\psi_{j,I}(t) = \sup_{z \in I} F_j(t, z). \tag{11}$$

Endpoint limits are included when the endpoint is infinite. Suppressing j, differentiation of (5) gives

$$\frac{\partial F(t,z)}{\partial z} = -\frac{s^2 + (t-m)(z-m)}{N\{s^2 + (z-m)^2/N\}^{3/2}}. \tag{12}$$

If $t > m$, the unique critical point is a maximum,

$$z^*(t) = m - \frac{s^2}{t-m}, \qquad F(t, z^*(t)) = \sqrt{\frac{(t-m)^2}{s^2} + \frac{1}{N}}. \tag{13}$$

If $t < m$, its critical point is a minimum; if $t = m$, the function is decreasing. Also $F(t, +\infty) = -1/\sqrt{N}$ and $F(t, -\infty) = 1/\sqrt{N}$. Thus the complete closed form is

$$\psi_{j,I}(t) = \max\left\{ F_j(t,\ell), F_j(t,u), \sqrt{\frac{(t-m_j)^2}{s_j^2} + \frac{1}{N}} \;\middle|\; t > m_j,\ z^*(t) \in I \right\}. \tag{14}$$

The conditional entry is omitted when its condition fails, rather than replaced by zero. Replacing it by zero would incorrectly inflate negative suprema.

The signed GWC score, quantile, and bounds are

$$\Phi_G(\mathbf{t}; x) = \max_j \psi_{j, D_j(x)}(t_j) = \sup_{\mathbf{z} \in D(x)} \Phi_{\mathbf{z}}(\mathbf{t}), \tag{15}$$

$$Q_G(x) = \mathcal{Q}_\alpha\big(\Phi_G(\mathbf{E}^1; x), \dots, \Phi_G(\mathbf{E}^n; x)\big), \tag{16}$$

$$U_j(x) = \omega_j(Q_G(x)), \quad G_j(x) = D_j(x) \cap (-\infty, U_j(x)], \quad G(x) = \prod_j G_j(x), \tag{17}$$

$$C_{\text{gwc}}(x) = \{y : \mathbf{E}(x,y) \in G(x)\}. \tag{18}$$

For a finite lower endpoint a_j, (14) uses $F_j(t, a_j)$, $-1/\sqrt{N}$, and the critical maximum when $t > m_j$ and $z^*(t) \geq a_j$. For $a_j = -\infty$ it simplifies to

$$\psi_{j,\mathbb{R}}(t) = \begin{cases} \sqrt{(t-m_j)^2/s_j^2 + 1/N}, & t > m_j, \\ 1/\sqrt{N}, & t \leq m_j. \end{cases}$$

Proposition 4.1 (GWC containment). *For every realized calibration sample and test input,*

$$C_{\text{full}}(x) \subseteq C_{\text{gwc}}(x).$$

Proof. For a candidate in $D(x)$, $\Phi_{\mathbf{e}}(\mathbf{E}^i) \leq \Phi_G(\mathbf{E}^i; x)$ for every i, whence $Q_{\mathbf{e}} \leq Q_G$. Full acceptance implies $g_j(e_j) \leq Q_{\mathbf{e}} \leq Q_G$, and Lemma 2.1 gives $e_j \leq U_j$. $\square$

The dependence of $D(x)$ on the test input causes no difficulty: this is a pathwise containment argument, not a claim that GWC calibration scores formed using that input are exchangeable on their own.

5 From full cells to surface envelopes

Assume $G(x)$ is nonempty. For each coordinate, partition G_j using its two endpoints and the distinct calibration residuals strictly between them:

$$a_j = \eta_{j,0} < \eta_{j,1} < \cdots < \eta_{j,K_j} = U_j, \qquad I_{j,k} = [\eta_{j,k-1}, \eta_{j,k}] \cap \mathbb{R}.$$

There are at most $n+1$ cells. We deliberately use closed cells, a cover with shared endpoints. This prevents boundary exclusions when residuals have atoms. A singleton G_j is represented by one singleton cell. Distinct interior knots avoid zero-width duplicate cells. The proofs allow any finite ordered interval cover, though Section 7.1 uses the calibration partition.

A full cell $\mathcal{I}_h = \prod_j I_{j,h_j}$ has exact local score

$$\Phi_h^{\text{exact}}(\mathbf{t}) = \sup_{\mathbf{z} \in \mathcal{I}_h} \Phi_{\mathbf{z}}(\mathbf{t}) = \max_j \psi_{j,I_{j,h_j}}(t_j).$$

Enumerating all h costs up to $(n+1)^d$ cells. Instead fix only the jth coordinate cell and let the others range over GWC:

$$H_j(\mathbf{t}) = \max_{\ell \neq j} \psi_{\ell,G_\ell}(t_\ell), \tag{19}$$

$$\overline{\Phi}_{j,k}(\mathbf{t}) = \max\{\psi_{j,I_{j,k}}(t_j), H_j(\mathbf{t})\} = \sup_{\substack{z_j \in I_{j,k} \\ z_\ell \in G_\ell,\ \ell \neq j}} \Phi_{\mathbf{z}}(\mathbf{t}), \tag{20}$$

$$\overline{Q}_{j,k} = \mathcal{Q}_\alpha\big(\overline{\Phi}_{j,k}(\mathbf{E}^1), \ldots, \overline{\Phi}_{j,k}(\mathbf{E}^n)\big). \tag{21}$$

Set $H_1 = -\infty$ for $d = 1$. Every transformation in these formulas is given explicitly by (14). For $h_j = k$ we have the pointwise chain

$$\Phi_{\mathbf{e}} \leq \Phi_h^{\text{exact}} \leq \overline{\Phi}_{j,k} \leq \Phi_G \quad \text{whenever } \mathbf{e} \in \mathcal{I}_h. \tag{22}$$

The envelope enlarges the candidate domain before taking the supremum. In general a quantile of pointwise maxima is not the maximum of the corresponding quantiles. Thus no equality with exact full-cell LWC is asserted.

Define the accepted coordinate piece and its upper endpoint by

$$A_{j,k} = I_{j,k} \cap (-\infty, \omega_j(\overline{Q}_{j,k})], \tag{23}$$

$$B_{j,k} = \begin{cases} \min\{\eta_{j,k}, \omega_j(\overline{Q}_{j,k})\}, & A_{j,k} \neq \varnothing, \\ -\infty, & A_{j,k} = \varnothing. \end{cases} \tag{24}$$

For a finite lower endpoint, nonemptiness is exactly $\omega_j(\overline{Q}_{j,k}) \geq \eta_{j,k-1}$; a value $-\infty$ never represents a finite residual. In particular, equality retains the singleton at the cell's left endpoint. Let

$$L_j = \max_{1 \leq k \leq K_j} B_{j,k}, \qquad R_{\mathrm{env}}(x) = D(x) \cap \prod_j (-\infty, L_j], \qquad C_{\mathrm{env}}(x) = \{y : \mathbf{E}(x,y) \in R_{\mathrm{env}}(x)\}. \tag{25}$$

If any coordinate has no accepted piece, return the empty set. Otherwise R_{env} is the coordinatewise lower enclosure of the pieces; it may fill gaps between them. It is not being identified with the exact local union.

Theorem 5.1 (Signed envelope validity)**.** *Pathwise,* $C_{\mathrm{full}}(x) \subseteq C_{\mathrm{env}}(x) \subseteq C_{\mathrm{gwc}}(x)$*. Consequently* $\Pr\{Y^{n+1} \in C_{\mathrm{env}}(X^{n+1})\} \geq 1 - \alpha$*.*

Proof. Take $y \in C_{\mathrm{full}}(x)$. Proposition 4.1 implies that its residual $\mathbf{e}$ belongs to $G(x)$. For each j, choose a covering cell $I_{j,k}$ containing e_j. Every calibration observation satisfies $\Phi_{\mathbf{e}}(\mathbf{E}^i) \leq \overline{\Phi}_{j,k}(\mathbf{E}^i)$. Quantile monotonicity and full acceptance therefore yield

$$g_j(e_j) \leq \max_\ell g_\ell(e_\ell) \leq Q_{\mathbf{e}} \leq \overline{Q}_{j,k}, \quad e_j \leq \omega_j(\overline{Q}_{j,k}), \quad e_j \in A_{j,k}, \quad e_j \leq B_{j,k} \leq L_j.$$

This proves the first containment simultaneously for all coordinates without a Bonferroni correction. Each nonempty $A_{j,k}$ lies in G_j, so $L_j \leq U_j$, which proves the second. The coverage statement follows from Theorem 3.1. □

6 Why it improves the old separated shortcut

Here the comparison uses the *same nonnegative calibration residuals*, test score function, level α, and calibration partition, with $D_j = [0, \infty)$. It does not compare signed residuals to capped or shifted residuals. Assume $s_j > 0$. Define the old GWC box G^{old} and

$$c_\ell = \inf_{z \in G_\ell^{\mathrm{old}}} \frac{\mu_\ell(z)}{\sigma_\ell(z)}, \qquad r_\ell(h) = \inf_{z \in I_{\ell,h}^{\mathrm{old}}} \sigma_\ell(z). \tag{26}$$

For nonnegative residuals $m_\ell \geq 0$ and $G_\ell^{\mathrm{old}} = [0, U_\ell^{\mathrm{old}}]$. Differentiating $\mu(z)/\sigma(z)$ gives

$$\frac{d}{dz} \frac{\mu(z)}{\sigma(z)} = \frac{s^2 - m(z - m)}{N \sigma(z)^3}.$$

Its interior critical point, when present, is a maximum. Thus

$$c_\ell = \min\left\{ \frac{\mu_\ell(0)}{\sigma_\ell(0)}, \frac{\mu_\ell(U_\ell^{\mathrm{old}})}{\sigma_\ell(U_\ell^{\mathrm{old}})} \right\},$$

with the second expression interpreted as $1/\sqrt{N}$ at infinity, and

$$r_\ell(h) = \sqrt{s_\ell^2 + \mathrm{dist}(m_\ell, I_{\ell,h}^{\mathrm{old}})^2 / N}.$$

The old separated score on a full cell is

$$\Phi_h^{\text{sep}}(\mathbf{t}) = \max_\ell \{t_\ell / r_\ell(h_\ell) - c_\ell\}, \qquad \mathbf{t} \geq 0.$$

If the old box contains the mean, choose each off-coordinate cell to contain m_ℓ. Its minimum scale is s_ℓ. The old mean-row score for surface (j,k) is consequently

$$\Phi_{j,k}^{\text{old}}(\mathbf{t}) = \max\left\{\frac{t_j}{r_j(k)} - c_j, \max_{\ell \neq j}\left(\frac{t_\ell}{s_\ell} - c_\ell\right)\right\}. \tag{27}$$

Its quantile and clipped coordinate pieces define L_j^{old} exactly as in (21)–(25), using the old cells. If the old mean cell is unavailable, the manuscript shortcut returns its GWC box.

Lemma 6.1 (Pointwise score domination)**.** *For $\mathbf{t} \geq 0$ and the same GWC box and cells, $\overline{\Phi}_{j,k}(\mathbf{t}) \leq \Phi_{j,k}^{\text{old}}(\mathbf{t})$.*

Proof. For $z \in I_{j,k}$, $\sigma_j(z) \geq r_j(k) > 0$ and $\mu_j(z)/\sigma_j(z) \geq c_j$. Nonnegativity gives

$$F_j(t_j, z) = \frac{t_j}{\sigma_j(z)} - \frac{\mu_j(z)}{\sigma_j(z)} \leq \frac{t_j}{r_j(k)} - c_j.$$

Taking the supremum proves the active-coordinate inequality. For $\ell \neq j$ and $z \in G_\ell$, $\sigma_\ell(z) \geq s_\ell$, so similarly $F_\ell(t_\ell, z) \leq t_\ell/s_\ell - c_\ell$. Take the off-coordinate suprema and then the maximum. Both steps explicitly use $t_\ell \geq 0$; reversing its sign reverses the comparison of the scale terms. □

Theorem 6.2 (Uniform weak improvement over the old shortcut)**.** *Under the preceding conditions, with closed-cell and weak-acceptance conventions for both constructions,*

$$C_{\text{env}}(x) \subseteq C_{\text{old}}(x), \qquad L_j \leq L_j^{\text{old}} \quad (j \leq d)$$

whenever both coordinate bounds represent nonempty sets. The set containment also holds when the envelope is empty or when the old shortcut falls back to GWC. In particular, each outcome interval and the resulting rectangular volume are weakly smaller, while finite-sample coverage remains at least $1 - \alpha$.

Proof. Lemma 6.1 and quantile monotonicity give $\overline{Q}_{j,k} \leq Q_{j,k}^{\text{old}}$. The signed link is increasing, so each accepted envelope piece is contained in its corresponding old piece. Taking coordinatewise suprema yields $L_j \leq L_j^{\text{old}}$. Taking score sublevel sets preserves containment in outcome space. On an old GWC fallback sample, Theorem 5.1 gives the result directly. □

The proof also works if the envelope's GWC box is a subset of the old box and its cells refine the old cells: each supremum is then over a smaller domain. Thus a harmless outward rounding in the old GWC computation does not reverse the mathematical comparison.

What the statement does and does not compare. The old manuscript uses some strict cell tests and assumes distinct residuals; the present note specifies a closed version to cover atoms as well. A literal comparison to code with a different tie policy requires numerical checking or harmonizing those policies; it is not justified by ignoring endpoints. The repository's old implementation also adds positive numerical jitter. The mathematical theorem is about the explicitly defined rules, not arbitrary floating-point perturbations. The paired experiment audit separately compares the actual implementations. During that audit, capped CQR scores exposed a specific bug in the former implementation: `argmax` selected the first tied zero below the mean, which could select a $[0,0]$ cell not containing the strictly positive mean. The off-coordinate scale then exceeded s_ℓ, so (27) no longer described that code. Selecting the last value at or below the mean (right-sided insertion rank) restores a mean-containing cell. All six saved failing examples satisfy containment after this correction; additional atom-heavy regression tests cover the same issue. Thus the theorem compares the mathematical old shortcut, including this necessary tie fix, and does not claim dominance over the buggy historical executable. The theorem is a containment result, not a claim that the envelope has higher coverage than a larger old set. Strict size improvement need not occur. Exact full-ratio cell LWC can be tighter than the surface envelope, and CQHR has a different scalarization; neither is dominated by this argument.

7 Computation and a justified search rule

Compute m, s and the GWC scores in $O(nd)$ operations. Sort the distinct calibration coordinates in $O(dn\log n)$. Precompute $V_{i\ell} = \psi_{\ell,G_\ell}(E^i_\ell)$ and all $H_{ij} = \max_{\ell\neq j} V_{i\ell}$ in $O(nd)$ using left and right cumulative maxima. Each surface quantile then requires $O(n)$ arithmetic and an order statistic selection, with no factor of d inside the surface search.

Lemma 7.1 (Rightmost surviving cell)**.** *For any ordered interval cover, the largest $B_{j,k}$ is the endpoint of the rightmost nonempty $A_{j,k}$. Thus scanning from K_j downwards and stopping at the first nonempty piece computes L_j exactly, without a monotone predicate.*

Proof. If cell k survives and $h < k$, then $B_{j,h} \leq \eta_{j,h} \leq \eta_{j,k-1} \leq B_{j,k}$ whenever h survives. An empty piece cannot increase the maximum. All cells to the right have already been checked and are empty. □

The worst-case cost is $O(dn^2 + dn\log n)$, and the actual search cost is $O(n\sum_j M_j)$ when M_j cells are inspected. The implementation records these counts. This proof does not claim that the values $B_{j,k}$ are monotone: they drop to the empty sentinel after a failure.

7.1 A range where binary search is proved

For $n > 1$ put $b_j = m_j - s_j/\sqrt{n-1}$. We prove a stronger search fact than the usual restriction to cells above the empirical mean, but make no unconditional prefix claim for the cells below b_j.

Lemma 7.2 (Rank-count predicate above b_j)**.** *Consider a calibration-partition cell $I_{j,k} = [\ell, u]$ with finite $\ell > b_j$ and $u > \ell$. Set $h_i = H_j(\mathbf{E}^i)$. Then*

$$A_{j,k} \neq \varnothing \iff \#\{i : E^i_j < \ell,\ h_i < g_j(\ell)\} < r. \tag{28}$$

Consequently the survival predicate is a true prefix followed by a false suffix on these cells, and binary search is valid on this part of the partition.

Proof. First $g_j(\ell) > -1/\sqrt{N}$ is equivalent to $\ell > b_j$ by (7). For fixed $t = \ell$, (12) shows that $F_j(\ell, z)$ has either no interior maximum or is decreasing for $z \geq \ell$. Its supremum over $[\ell, \infty)$ is the larger of $g_j(\ell)$ and $-1/\sqrt{N}$, hence is $g_j(\ell)$. More explicitly, if $\ell < m_j$ the interior critical point is a minimum; if $\ell \geq m_j$ the derivative is negative on this half-line.

If $t < \ell$, then $F_j(t, z) < F_j(\ell, z)$ at every finite z, and its limit at infinity is strictly below $g_j(\ell)$. Continuity, including that limit, therefore implies $\psi_{j,I_{j,k}}(t) < g_j(\ell)$. If $t = \ell$, the supremum equals $g_j(\ell)$. If $t > \ell$, evaluating at $z = \ell$ already gives a value strictly larger than $g_j(\ell)$. It follows that the ith surface score is strictly below $g_j(\ell)$ exactly when both inequalities inside the count in (28) hold. Its rth order statistic is at least $g_j(\ell)$ exactly when fewer than r scores are strictly below this value. Link inversion proves the equivalence. As ℓ increases, $g_j(\ell)$ increases, so the counted sets are nested and their cardinalities cannot decrease. □

In fact, the proof of the rank predicate uses only $z \geq \ell$ and the left endpoint; no sign assumption is placed on the calibration residuals. One can check the top cell first, use binary search on the cells with left endpoint above b_j, and scan backwards below that range only if no upper cell survives. This has $O(dn \log n)$ search cost when every coordinate boundary is found in the proved range, with the unconditional backward bound otherwise. Observed prefix behavior outside the proved range must not be substituted for a proof. The initial reference implementation uses unconditional backward search so its correctness does not depend on this optional acceleration.

7.2 Direct order-statistic localization

Binary search is not necessary in the certified range. For each coordinate set

$$W_{ij} = \max\{E^i_j,\, \omega_j(H_j(\mathbf{E}^i))\}, \qquad T_j = W_{(r),j},$$

where $W_{(r),j}$ is the rth order statistic, with the same infinite-rank convention. The inverse is evaluated separately at each off-coordinate score; it is not the inverse of an aggregate quantile.

Proposition 7.3 (One-order-statistic localization)**.** *For a nondegenerate cell with left endpoint* $\ell > b_j$*,*

$$A_{j,k} \neq \varnothing \iff \ell \leq T_j.$$

In particular every cell with left endpoint exceeding $\max\{b_j, T_j\}$ *can be discarded without evaluating a surface quantile.*

Proof. The extended inverse convention gives $h_i < g_j(\ell) \iff \omega_j(h_i) < \ell$ even when h_i is outside the open range of g_j. Hence

$$\{E^i_j < \ell,\ h_i < g_j(\ell)\} = \{\max(E^i_j, \omega_j(h_i)) < \ell\} = \{W_{ij} < \ell\}.$$

There are fewer than r such values exactly when $W_{(r),j} \geq \ell$. Apply Lemma 7.2. The discarded cells are all in that lemma's range, so this pruning makes no claim about the remaining lower cells. □

Select T_j in $O(n)$ time and find the last partition left endpoint at or below $\max\{b_j, T_j\}$ by insertion search in $O(\log n)$ time. Start the unconditional backward scan there. If this cell has left endpoint above b_j, it is the rightmost survivor and only its surface quantile is needed to compute the actual clipped endpoint $B_{j,k}$. The endpoint is generally *not* T_j: the order statistic localizes the cell, after which (21)–(25) still determine its boundary. If the candidate cell is below the certified range, continue backward normally. Thus search costs $O(nd + d \log n)$ when all coordinate boundaries lie in the certified range, after sorting and off-coordinate preprocessing, with the unconditional worst-case bound otherwise. The optional implementation `search="rank"` uses this pruning and an outward numerical guard near ties. It need not be faster on typical samples: a backward scan that already checks one cell avoids the extra transformed order statistic.

8 Outcome intervals, signed improvements, and edge cases

For absolute residuals $E_j(x,y) = |y - \hat{f}_j(x)|$, $a_j = 0$ and the prediction interval is $[\hat{f}_j(x) - L_j, \hat{f}_j(x) + L_j]$. For ordered fitted quantiles $q_j^-(x) \le q_j^+(x)$, signed CQR scores satisfy

$$\begin{aligned} E_j(x,y) &= \max\{q_j^-(x) - y,\ y - q_j^+(x)\} \\ &= \left| y - \frac{q_j^-(x) + q_j^+(x)}{2} \right| - \frac{q_j^+(x) - q_j^-(x)}{2}, \qquad a_j(x) = -\frac{q_j^+(x) - q_j^-(x)}{2}. \end{aligned}$$

Consequently

$$E_j(x,y) \le L_j \iff q_j^-(x) - L_j \le y \le q_j^+(x) + L_j, \qquad \text{length}_j = (q_j^+ - q_j^-) + 2L_j.$$

A negative L_j shrinks the base interval. A bound below a_j means an empty interval, not a singleton created by clamping. A bound equal to a_j gives the singleton at the midpoint. Mixed signs across coordinates are allowed; GWC inflation in one or all coordinates does not rule out local shrinkage. The envelope's computations for one coordinate still depend on the other coordinates through H_j, so a general sign-reflection shortcut is not implied.

Small calibration samples. If $r = N$, all conformal quantiles are $+\infty$ and the procedure returns the entire attainable domain. An implementation must preserve this case, not replace the quantile by the largest observed score.

Zero calibration variance. To extend the definition to all finite calibration samples, define the full conformal reference scale to be the augmented sample standard deviation when positive, and 1 when the entire augmented coordinate is constant. This rule is symmetric, so Theorem 3.1 still holds. If any observed $s_j = 0$, the reference implementation returns the whole attainable domain, a valid superset of that same reference set. Otherwise all proofs above apply. This conservative fallback avoids division by zero; it can be costly for heavily capped scores. The positive-score dominance theorem excludes these samples because the unregularized old shortcut also lacks a positive scale. A practical regularized variant could instead add a predetermined positive ridge to every augmented variance symmetrically, but its comparison must then be made to an old shortcut with matching regularization.

Infinity, empty boxes, and ties. If GWC is empty, return empty immediately. Infinite cell endpoints use the limits in (14). An infinite upper coordinate makes volume infinite unless some coordinate interval is empty or has zero length, in which case rectangular Lebesgue volume is zero. Closed cells and weak inequalities are maintained across calibration ties and candidate ties. The finite-sample coverage guarantee is marginal over the calibration/test experiment; it is not a guarantee conditional on the realized calibration sample or on every x.

9 Reproducibility and comparison protocol

The implementation is `utility/envelope.py`. The numerical formula and containment audit is `envelope_method/verify.py`. Its machine-readable results are in `envelope_method/verification.json`. Experimental workloads, their configurations, source data hashes, trial data, and completion states are recorded under `envelope_method/`. Every synthetic empirical trial generates fresh training, calibration, and test observations and fits a new prediction model. Compared methods share those fitted predictions and residuals within a trial. Only the real-data studies resplit a fixed observed dataset. The residual-array experiments used to audit algebra and search rules are mathematical checks, not estimates of a fitted model's coverage. Full conformal coverage is established by the proofs, not inferred from Monte Carlo results.

For a nonnegative comparison use the same residuals for both algorithms. For signed CQR comparisons retain raw residuals and measure outcome-space lengths and volumes, including empty intervals. Applying the envelope also to capped and shifted residuals separately identifies the effect of improving the local construction versus changing the score representation. Base interval miscoverage 0.1 means fitted marginal quantiles 0.05 and 0.95; it is different from the simultaneous target miscoverage, even when both are 0.1. Use the same base intervals for envelope and CQHR in each such comparison. No uniform envelope–CQHR ordering is claimed. Store trial-level results and compute uncertainty over independent trials, not by treating all test points from the same fitted/calibrated split as independent trials.

References

[1] G. Shafer and V. Vovk (2008). A Tutorial on Conformal Prediction. *Journal of Machine Learning Research* 9:371–421. `https://jmlr.org/papers/v9/shafer08a.html`.